

TAMAM WEBSITE

DigitalOcean — Server Setup

2 vCPU | 8 GB RAM | 160 GB SSD | Ubuntu 22.04

Field	Details
Date	13 April 2026
Droplet Plan	Basic — 2 vCPU / 8 GB RAM / 160 GB SSD
OS	Ubuntu 22.04 LTS
Web Server	Apache 2.4
PHP	PHP 8.1
Database	MySQL 8.0
Estimated Cost	~\$48 / month

1. Create the Droplet

Go to cloud.digitalocean.com > Create > Droplets and use these settings:

Setting	Value
Image	Ubuntu 22.04 LTS x64
Plan	Basic — 2 vCPU / 8 GB RAM / 160 GB SSD
Region	Closest to your users
Authentication	SSH Key
Hostname	tamam-website

2. Firewall

Go to Networking > Firewalls > Create Firewall and assign it to the Droplet.

Port	Allow From
------	------------

22 (SSH)	Your IP only or Anywhere
80 (HTTP)	Anywhere
443 (HTTPS)	Anywhere
3306 (MySQL)	Block — no rule

3. Update System

```
apt update && apt upgrade -y
apt install -y curl wget unzip git software-properties-common
```

4. Install Apache

```
apt install apache2 -y
a2enmod rewrite ssl headers
systemctl enable apache2
systemctl start apache2
```

5. Install PHP 8.1

```
add-apt-repository ppa:ondrej/php -y
apt update

apt install -y \
  php8.1 libapache2-mod-php8.1 \
  php8.1-mysql php8.1-curl php8.1-gd \
  php8.1-mbstring php8.1-xml php8.1-zip \
  php8.1-intl php8.1-soap php8.1-imap \
  php8.1-bcmath php8.1-openssl php8.1-sodium

php -v
```

6. Install MySQL 8.0

```
apt install mysql-server -y
systemctl enable mysql
systemctl start mysql
mysql_secure_installation
```

7. Install Composer

```
curl -sS https://getcomposer.org/installer | php
mv composer.phar /usr/local/bin/composer
chmod +x /usr/local/bin/composer
```

8. Install Git

```
apt install git -y
git --version
```

9. Install SSL (Certbot)

```
apt install certbot python3-certbot-apache -y
```

10. Security

```
# SSH - disable password login
nano /etc/ssh/sshd_config
# Set: PasswordAuthentication no
# Set: PermitRootLogin no
systemctl restart ssh

# Firewall
apt install ufw -y
ufw default deny incoming
ufw default allow outgoing
ufw allow 22 && ufw allow 80 && ufw allow 443
ufw enable

# Fail2Ban
apt install fail2ban -y
systemctl enable fail2ban
systemctl start fail2ban

# PHP - hide version, disable error display
# Edit /etc/php/8.1/apache2/php.ini
# expose_php = Off
# display_errors = Off
systemctl restart apache2
```

11. GitHub Actions Runner

Go to GitHub repo > Settings > Actions > Runners > New self-hosted runner > Linux and follow the commands shown.

```
# After running config.sh from GitHub:
sudo ./svc.sh install
```

```
sudo ./svc.sh start
```

12. Non-Root User Setup

A dedicated deploy user will be created with sudo privileges.

- SSH key-based authentication will be configured for this user.
- Root login will be used only for initial setup and then avoided entirely.

Steps:

- Create user: `adduser deploy`
- Add to sudo group: `usermod -aG sudo deploy`
- Configure SSH access via authorized keys
- Disable password authentication (optional but recommended)

13. Deployment Flow

- A GitHub Actions self-hosted runner will run on the server under the `deploy` user.
- On each push:
 - The runner pulls the latest code into the respective directory.
 - Dependencies (optional: `composer install`) can be executed automatically.
 - Web server (Apache) is restarted to reflect changes.

Flow Summary:

1. Developer pushes code to GitHub branch
2. GitHub Actions triggers workflow
3. Self-hosted runner executes on server
4. Code is pulled into correct directory
5. (Optional) Run `composer install`
6. Restart Apache service

14. Environment Structure

The server will maintain separate directories for each environment:

- 1 - /var/www/dev/ → dev.tamam.website (dev branch)
- 2 - /var/www/sub/ → sub.tamam.website (stage branch)
- 3 - /var/www/html/ → tamam.website (main/production branch)

- Each environment is mapped to a corresponding domain/subdomain.
- Git branches are aligned with environments:
 - `dev` → development
 - `stage` → staging (sub)
 - `main` → production

15. Database Setup

- A dedicated database and user will be created.

Configuration:

- Database: `tamam_db`
- User: `tamam_user`
- Host restriction: `localhost` only

Security measures:

- MySQL bound to `127.0.0.1` (no external access)
- Least privilege access granted to the database user

Steps:

```
CREATE DATABASE tamam_db;  
CREATE USER 'tamam_user'@'localhost' IDENTIFIED BY 'strong_password';  
GRANT ALL PRIVILEGES ON tamam_db.* TO 'tamam_user'@'localhost';  
FLUSH PRIVILEGES;
```

16. Backup Plan (Recommendation)

Since the database is hosted on the same server (no managed DB), backups are critical to prevent data loss.

Recommended Minimum Setup

1. Daily Backup (Free)

- Use `mysqldump` via cron job
- Stores backups locally or optionally uploads to remote storage

Example cron:

```
0 2 * * * mysqldump -u tamam_user -p'password' tamam_db > /backup/tamam_db_$(date +%F).sql
```

2. Droplet Snapshots (Optional but Recommended)

- Weekly snapshots of the entire server
- Approximate cost: ~\$10/month

16. Estimated Monthly Cost

Item	Cost
Droplet — 2 vCPU / 8 GB RAM / 160 GB SSD	~\$48 / month
SSL — Let's Encrypt	Free
Firewall	Free
DNS	Free
Bandwidth — 5 TB included	Free
Optional: Weekly Droplet Snapshots	+~\$10 / month
TOTAL	~\$48 / month

17. Checklist

- ☐ Create Droplet — Ubuntu 22.04, 8 GB RAM, 160 GB SSD
- ☐ Set up DigitalOcean Firewall — allow 80, 443, 22 from your IP only
- ☐ apt update && apt upgrade
- ☐ Install Apache 2.4
- ☐ Install PHP 8.1 and all extensions
- ☐ Install MySQL 8.0 and run mysql_secure_installation
- ☐ Install Composer
- ☐ Install Git
- ☐ Install Certbot
- ☐ Disable SSH password login
- ☐ Install and enable ufw
- ☐ Install and start Fail2Ban
- ☐ Set expose_php = Off and display_errors = Off in php.ini
- ☐ Register GitHub Actions self-hosted runner